Heine–Borel Theorem

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

A complete authored walkthrough of heine–borel theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RA-02 / Real Analysis / Report v1

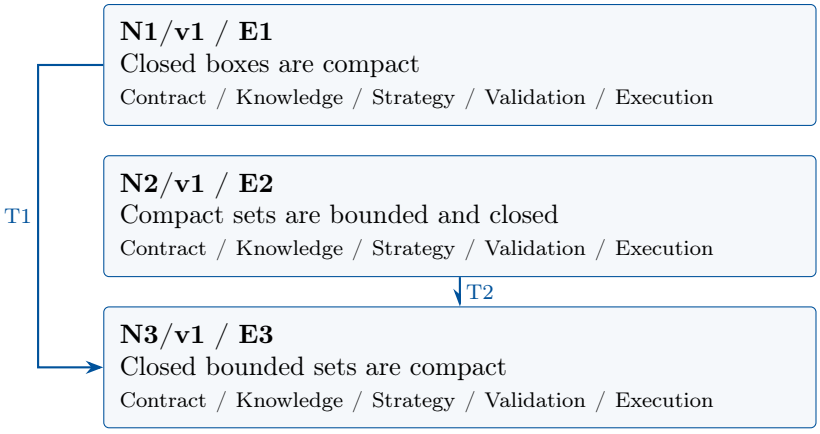
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For $K \subset \mathbb{R}^n$ with its Euclidean topology, prove that K is compact if and only if it is closed and bounded.*

A. Foundations and permitted imports

Use completeness of R to prove compactness of closed boxes below, the Euclidean distance, and finite-volume geometry. Compact means every open cover has a finite subcover. The proof supplies the needed closed-box lemma rather than importing Heine–Borel itself.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N3. Every closed bounded box is compact.

Receiving activity. N3: Closed bounded sets are compact

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Closed bounded K; compact box; prior exports E1/v1, E2/v1

T2: N2/E2/v1 to N3. Compact K is closed and bounded.

Receiving activity. N3: Closed bounded sets are compact

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Closed bounded K; compact box; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Closed boxes are compact

Object and role. Use nested bisection and completeness.

Inputs. An open cover of a closed box in R^n; foundational imports stated in §1

C — Contract

Own E1/v1: Every closed bounded box is compact.

K — Knowledge

Retain An open cover of a closed box in R^n; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Use nested bisection and completeness.

V — Validation

The cover sets are open in an ambient neighborhood; nested endpoints must be retained.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Suppose a box Q has an open cover with no finite subcover.

N1.2 Bisect each side; at least one of the 2^n closed subboxes has no finite subcover, since otherwise their finite covers unite.

N1.3 Continue to nested boxes Q_j whose diameters tend to 0.

N1.4 Completeness in each coordinate supplies a common point x; it is unique by the diameter bound.

N1.5 A covering open set contains a ball $B(x, \varepsilon)$.

N1.6 For large j every point of Q_j is within its diameter of x, so Q_j lies in that one set, contradiction.

N1.7 Degenerate sides cause no difficulty.

Export E1/v1. Every closed bounded box is compact.

Acceptance gate. The cover sets are open in an ambient neighborhood; nested endpoints must be retained. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: The cover sets are open in an ambient neighborhood; nested endpoints must be retained. Recheck dependent nodes N3.

B. N2/v1: Compact sets are bounded and closed

Object and role. Use a ball cover and point separation.

Inputs. Compact K and Euclidean Hausdorff geometry; foundational imports stated in §1

C — Contract

Own E2/v1: Compact K is closed and bounded.

K — Knowledge

Retain Compact K and Euclidean Hausdorff geometry; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Use a ball cover and point separation.

V — Validation

Boundedness alone does not supply a finite subcover.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 The increasing balls $B(0, j)$, $j \geq 1$, cover K; a finite subcover places it in one sufficiently large ball.

N2.2 For $x \notin K$ and each $y \in K$ choose disjoint balls U_y around y and V_y around x, using distance $(x, y) > 0$.

N2.3 Finitely many U_y cover K.

N2.4 The intersection of the corresponding V_y is an open neighborhood of x disjoint from K.

N2.5 Thus the complement of K is open and K is closed.

N2.6 The empty set satisfies both conclusions directly.

Export E2/v1. Compact K is closed and bounded.

Acceptance gate. Boundedness alone does not supply a finite subcover. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Boundedness alone does not supply a finite subcover. Recheck dependent nodes N3.

C. N3/v1: Closed bounded sets are compact

Object and role. Adjoin the complement to an open cover.

Inputs. Closed bounded K; compact box; prior exports E1/v1, E2/v1

C — Contract

Own E3/v1: $K \subset R^n$ is compact iff closed and bounded.

K — Knowledge

Retain Closed bounded K; compact box; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

S — Strategy

Adjoin the complement to an open cover.

V — Validation

Relative open sets must be converted correctly to a cover of Q.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Place K in a closed box Q.

N3.2 Given a relative open cover of K, represent its members as $K \cap O_i$ with ambient O_i open.

N3.3 The O_i together with $R^n \setminus K$ cover Q.

N3.4 Compactness of Q yields finitely many sets; discarding the complement leaves finitely many O_i covering K.

N3.5 This proves compactness and joins both directions of the equivalence.

Export E3/v1. $K \subset R^n$ is compact iff closed and bounded.

Acceptance gate. Relative open sets must be converted correctly to a cover of Q. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Relative open sets must be converted correctly to a cover of Q. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Closed boxes are compact

1. Suppose a box Q has an open cover with no finite subcover.
2. Bisect each side; at least one of the 2^n closed subboxes has no finite subcover, since otherwise their finite covers unite.
3. Continue to nested boxes Q_j whose diameters tend to 0.
4. Completeness in each coordinate supplies a common point x; it is unique by the diameter bound.
5. A covering open set contains a ball $B(x, \varepsilon)$.
6. For large j every point of Q_j is within its diameter of x, so Q_j lies in that one set, contradiction.
7. Degenerate sides cause no difficulty.

B. §4.N2 — Compact sets are bounded and closed

1. The increasing balls $B(0, j)$, $j \geq 1$, cover K; a finite subcover places it in one sufficiently large ball.
2. For $x \notin K$ and each $y \in K$ choose disjoint balls U_y around y and V_y around x, using distance $(x, y) > 0$.
3. Finitely many U_y cover K.
4. The intersection of the corresponding V_y is an open neighborhood of x disjoint from K.
5. Thus the complement of K is open and K is closed.
6. The empty set satisfies both conclusions directly.

C. §4.N3 — Closed bounded sets are compact

1. Place K in a closed box Q.
2. Given a relative open cover of K, represent its members as $K \cap O_i$ with ambient O_i open.
3. The O_i together with $R^n \setminus K$ cover Q.

4. Compactness of Q yields finitely many sets; discarding the complement leaves finitely many O_i covering K.

5. This proves compactness and joins both directions of the equivalence.

□

V. WORKED ORCHESTRATION

For $K = [0,1]^2$, the nested-box proof applies to every open cover, not just a grid. In contrast (0,1) is covered by set $s(1/j, 1), j \geq 2$ with no finite subcover; it is bounded but not closed in R.

A. Boundary and hypothesis audit

The unit coordinate vectors e_j in the closed unit ball of ℓ^2 have mutual distance $\sqrt{2}$, so there is no convergent subsequence. The infinite-dimensional analogue of the box argument has no $finite 2^n$ subdivision and fails. Validation preserves the finite-dimensional Euclidean scope.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.

6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

The unit coordinate vectors e_j in the closed unit ball of ℓ^2 have mutual distance $\sqrt{2}$, so there is no convergent subsequence. The infinite-dimensional analogue of the box argument has no $finite 2^n$ subdivision and fails. Validation preserves the finite-dimensional Euclidean scope.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.